

RIDHAM TANEJA

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SKILLS

Languages: Python

AI & ML: LangChain, RAG, Prompt Engineering, Multi-Agent Systems, XGBoost, LightGBM, AdaBoost, CNN, LSTM, BiLSTM, Attention Mechanisms

Frameworks & Libraries: FastAPI, Streamlit, Scikit-learn, TensorFlow, Keras, NumPy, Pandas, Matplotlib

Backend & Databases: SQL, REST APIs, FastAPI, Webhooks, Event-Driven Workflows, SQLite, ChromaDB, PostgreSQL

Deployment & DevOps: Docker, Hugging Face Spaces, Ollama, Git, GitHub, Jupyter Notebook

Concepts: Sequence Modelling, Class Imbalance Handling, Hyperparameter Tuning, DBMS, Cross-Validation, ROC-AUC, Precision-Recall, OOP

PROJECTS

UPI Sentinel · [GitHub](#) | [Live Demo](#) ↗

- Developed a contextual fraud detection system using CNN + BiLSTM + Attention to model transaction sequences and capture temporal spending behaviour.
- Addressed extreme class imbalance in 1.2M+ digital payment transactions through class weighting and threshold optimisation, achieving **0.85 F1-score**, **0.99 ROC-AUC**, and **90% recall**.
- Built a real-time inference pipeline using FastAPI and Streamlit, enabling sequence-aware fraud scoring through REST APIs and an interactive monitoring dashboard.
- Containerised and deployed** the complete application using **Docker and Hugging Face Spaces**, making the system publicly accessible for live demonstrations and evaluation.

GuestFlow-AI · [GitHub](#)

- Built an event-driven hospitality workflow automation platform using FastAPI, Streamlit, SQLite, and LangChain-compatible architecture, simulating real-world hotel CRM and guest support operations.
- Developed a **multi-agent orchestration system** that dynamically routes guest events across Pre-Stay, In-Stay, and Post-Stay workflows with escalation handling, VIP prioritisation, and persistent guest state management.
- Implemented **Retrieval-Augmented Generation (RAG)** with local LLM inference via Ollama (Phi-3), enabling context-aware guest query resolution using a hotel knowledge base.
- Designed a modern operations dashboard with real-time workflow visualisation, webhook-triggered automation, API integrations, ticket generation, and workflow analytics for hospitality support automation.

EXPERIENCE

AI/ML Trainee & Project Intern

October 2024 – September 2025

[QMANSYS INFOSOLUTIONS](#) (in association with [NASSCOM](#))

- Participated in an industry-oriented AI/ML training and project collaboration program involving practical implementation tasks and real-world technical assignments.
- Engineered a multi-class emotion classification system on 111K samples, achieving **96% accuracy** using XGBoost and **94%** with LightGBM.
- Improved model performance using hyperparameter tuning, cross-validation, ROC-AUC, and Precision-Recall metrics, achieving **95% validation accuracy** before deployment via Streamlit for real-time inference.

EDUCATION

B.Tech – Computer Science & Engineering (AI & ML)

Sep 2022 – Present

[Raj Kumar Goel Institute of Technology, Ghaziabad, Uttar Pradesh](#)

CGPA: **7.23 / 10**

Class 12 — CBSE Board

July 2021

[Harmilap Mission School, Kanpur, Uttar Pradesh](#)

Percentage: **76%**

Class 10 — CBSE Board

July 2019

[St. Xavier's High School, Vapi, Gujarat](#)

Percentage: **82%**